NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY

9648/03

Paper 3 Applications and Planning

September 2015

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do no use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Answer questions **1** to **4** in the spaces provided on the question paper.

Answer question **5** on the separate answer paper provided.

For Examiner's Use	
1	
2	
3	
5	
Total	60
For Examiner's Use	
4	12

This document consists of **18** printed pages and **0** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

1 **Fig. 1.1** outlines how a gene coding for human insulin is produced by genetic engineering techniques.

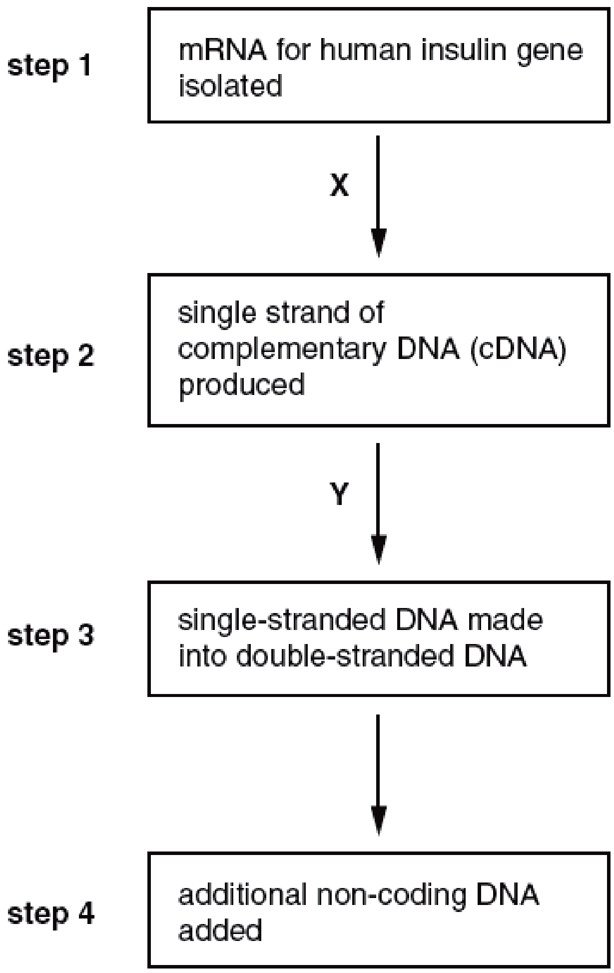

Fig. 1.1

(a) **(i)** Name the enzymes X and Y.

X ....

Y ....

[2]

(ii) Explain why the starting point in this procedure is mRNA.

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[2]

(iii) Suggest why some methods of manufacturing genetically engineered insulin use eukaryotic yeast cells rather than prokaryotic bacterial cells.

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[2]

(b) The artificial plasmid, pBR322, was constructed to act as a vector. It has often been used to insert human genes, such as the human insulin gene, into the bacterium, *Escherichia coli*.

The plasmid was constructed to include two genes, each giving resistance to a different antibiotic: an ampicillin resistance gene and a tetracycline resistance gene. The plasmid also has a target site for the restriction enzyme, *Bam*HI, in the middle of the tetracycline resistance gene.

A pBR322 plasmid was cut using *Bam*HI and the cDNA gene for human insulin inserted into it.

Fig. 1.2 shows pBR322 and the recombinant plasmid.

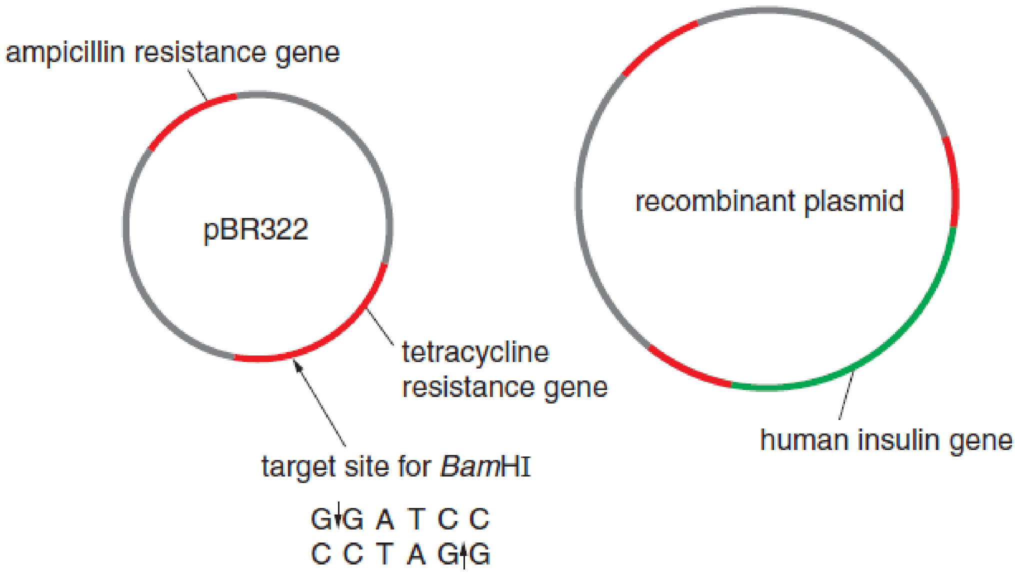

Fig. 1.2

With reference to **Fig. 1.2**, describe how a cDNA human insulin gene can be inserted into pBR322 that has been cut by *Bam*HI.

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[3]

(c) Bacteria were then mixed with the recombinant plasmids. Those bacteria which had successfully taken up recombinant plasmids were identified using the following steps:

step 1 – the bacteria were spread onto culture plates containing nutrient agar and ampicillin and incubated to allow colonies to form

step 2 – some bacteria from each of the colonies growing on these plates were transferred to plates containing nutrient agar and tetracycline, as shown in **Fig. 1.3**.

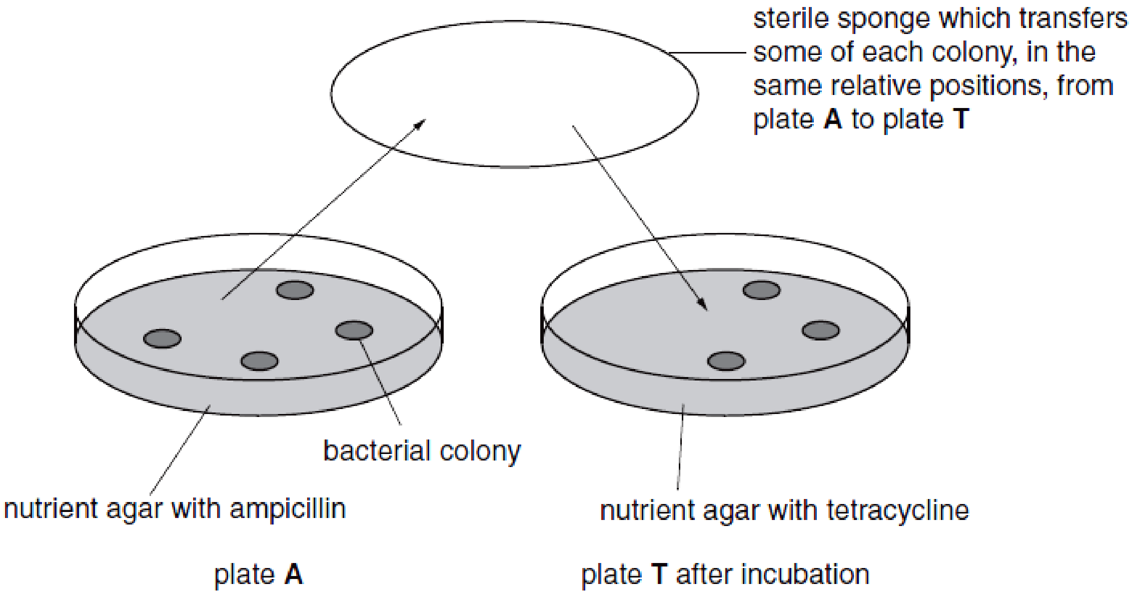

Fig. 1.3

(i) Explain why the bacteria were first spread onto plates containing ampicillin.

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[2]

(ii) Explain why it is important, for identifying bacteria that have successfully taken up the recombinant plasmid, that on pBR322 the target site for *Bam*HI is in the middle of the tetracycline resistance gene.

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[3]

(iii) Use a label line and the letter **C** to identify, on **Fig. 1.3**, a colony of bacteria that contain the recombinant plasmid.

[1]

[Total: 15]

2 **Fig. 2.1** shows the inheritance of SCID in a family in which both parents are carriers. Only one of the sons suffers from SCID.

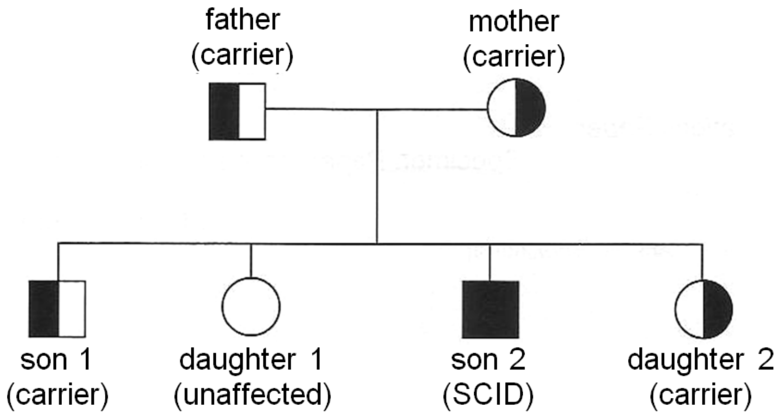


Fig. 2.1

Two RFLP morphs have been found tightly linked to the adenosine deaminase (ADA) gene which can be used for genetic screening. A radioactive probe has been designed to reveal the two RFLP morphs.

Both RFLP morphs and radioactive probe are shown in **Fig. 2.2** with the arrows representing the *Pvu*II restriction sites. The asterisk (*) indicates the position of a mutation at the restriction site that results in the loss of the restriction site. Morph 2 was found to be linked to the mutant ADA gene.

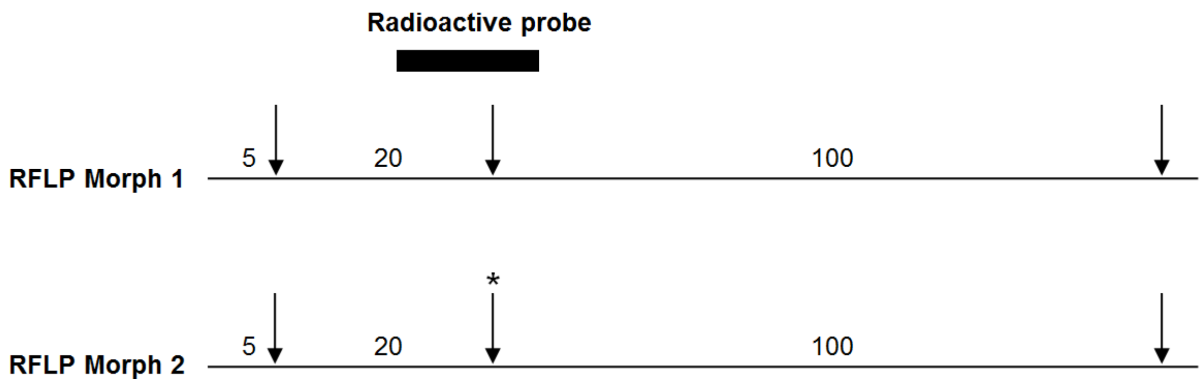
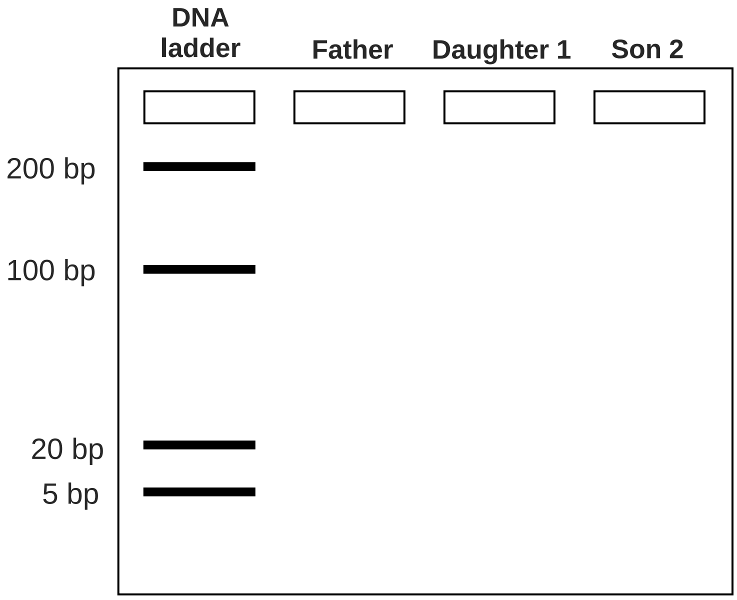


Fig. 2.2

(a) **(i)** With reference to **Fig 2.1** and **Fig. 2.2**, draw the resulting autoradiogram you would expect to see in the space below.



[3]

(ii) Explain the role of the DNA ladder.

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[1]

(b) Gene therapy can be used to treat severe combined immunodeficiency disease (SCID). A deficiency of adenosine deaminase (ADA) is a cause of severe combined immunodeficiency syndrome (SCID). Affected individuals are unable to produce ADA. Without this enzyme, T lymphocytes (a type of white blood cell) cannot provide immunity to infection. **Fig. 2.3** shows the processes involved in the treatment of ADA deficiency by gene therapy.

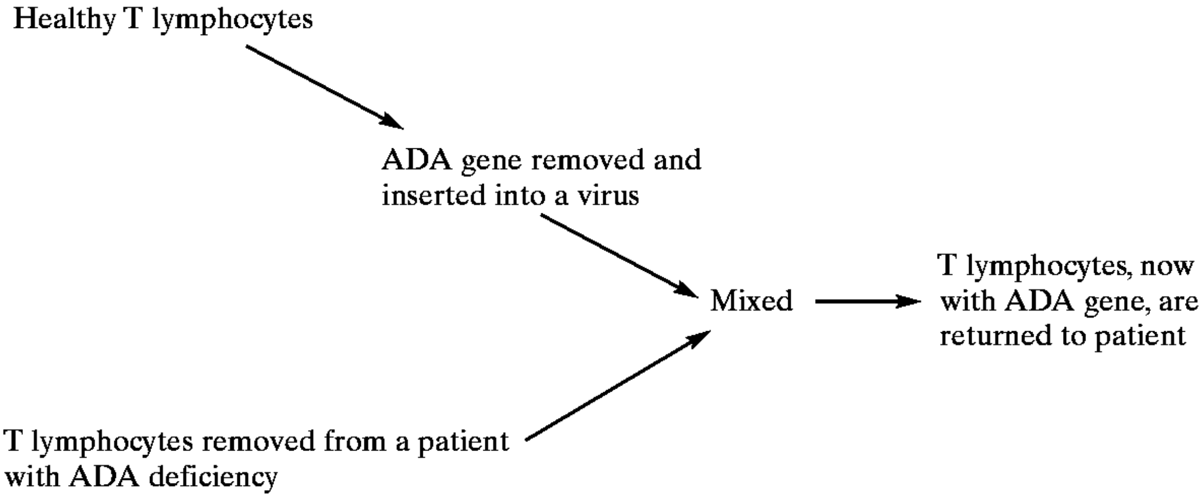

Fig. 2.3

(i) Explain why individuals who have been treated by this method of gene therapy do not pass on the functional ADA allele to their children.

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[2]

(ii) Suggest the advantages of a viral gene delivery system.

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[2]

(iii) Suggest why the ex vivo method was used in this case.

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[1]

(iv) Other than the short-lived nature of the treatment, explain other problems encountered in using gene therapy as a treatment for genetic diseases, such as SCID.

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[3]

(v) Name another method, other than gene therapy, which can be used for the treatment of ADA-deficiency.

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[1]

[Total: 13]

3 Plant tissue culture has many commercial applications. These include the production of large numbers of genetically identical plants and the culturing of plant cells in a bio-reactor for the production of secondary products.

Early attempts to culture plant tissue on a culture medium were not always successful.

The culture medium contained a range of substances that were known to be requirements for plant cell growth and division. These included:

- elements, present as mineral ions
- a sugar suitable for plant cells to use

(a) Explain why a sugar was included in the plant tissue culture medium.

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[2]

Investigations have shown that the original culture medium lacked key components called plant growth regulators needed for successful plant cell growth and division.

Although early attempts to culture plant tissue were not always successful, some investigators have reported successful growth using root tips.

(b) Suggest one reason why plant tissue culture was more successful using organs such as root tips.

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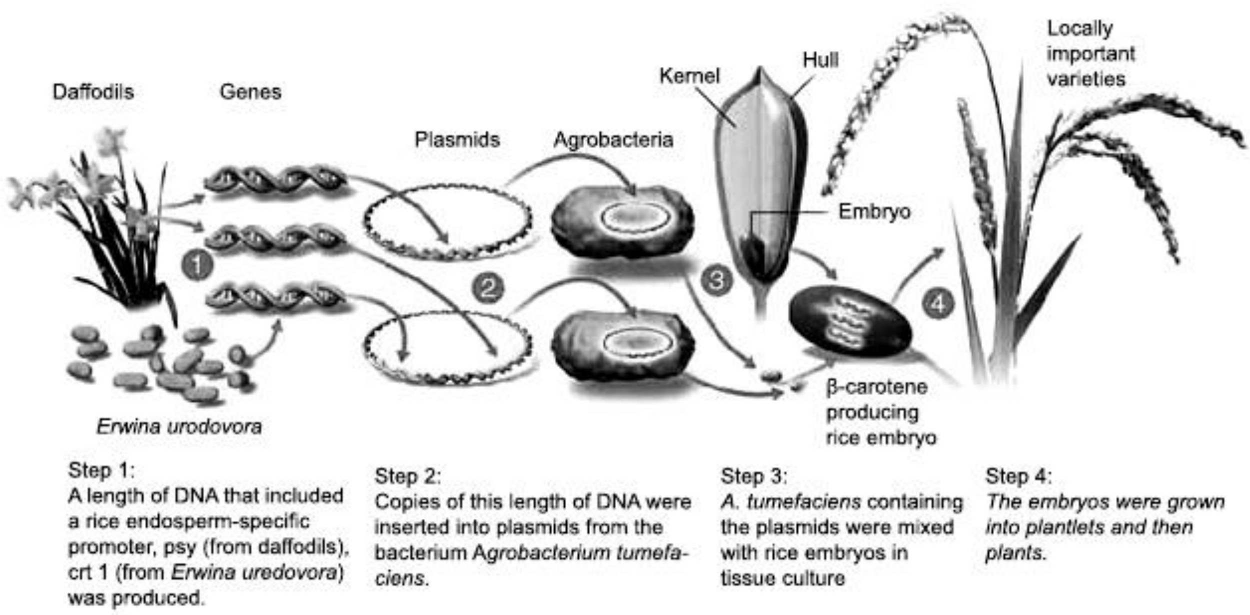
....

[1]

(c) Rice plants produce the precursor of vitamin A, β-carotene, in their green tissues, but not in the edible endosperms of their seeds because the seeds lack the enzymes for two steps of the pathway for β-carotene production. Genetic engineering methods and plant tissue culture techniques have been used to insert the genes for these enzymes into the rice embryos to produce rice seeds that contain β-carotene. These rice seeds are known as Golden Rice™. The production of such rice is an important step towards alleviating vitamin A deficiency in many rice-based societies in South-east Asia.



(i) Explain why *Agrobacterium tumefaciens* containing plasmids were used in Step 3.

....

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[2]

(ii) Besides Agrobacterium-mediated gene transfer, state one method that is commonly used to insert DNA into cells of monocotyledonous plants.

....
[1]

(d) The concentration of β-carotene in the seeds from the field trial of the original strain of Golden Rice™ in 2004 was only about 6μgg^{-1}. The rate limiting step in the production of β-carotene was found to be the activity of the enzyme encoded by the *psy* gene. Suggest what is meant by a *rate limiting step* in a metabolic pathway.

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[2]

(e) The concentration of β-carotene was increased by breeding with other varieties of rice and by further genetic engineering. More recent strains, such as Golden Rice 2™, contain about 31μgg^{-1} of β-carotene in the endosperm of the seeds.

Suggest what further genetic engineering was carried out to increase the concentration of β-carotene in the seeds.

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[2]

(f) Field trials of genetically engineered crops, such as Golden Rice™, hope to identify risks to the environment or to human health of growing and eating the crops.

Suggest

(i) one possible risk to the environment of growing a genetically engineered crop.

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[1]

(ii) one possible risk to human health of eating a genetically engineered crop.

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[1]

[Total: 12]

Planning Question

4 You are required to plan an investigation to determine how the age of the mealworms affect their rates of respiration.

A bicarbonate indicator is a type of pH indicator that is sensitive enough to show a colour change as the concentration of carbon dioxide gas in an aqueous solution changes.

increasing CO_2 in indicator ←			atmospheric CO_2 level			decreasing CO_2 in indicator →		
yellow		orange		red		magenta		purple
pH 7.6	pH 7.8	pH 8.0	pH 8.2	pH 8.4	pH 8.6	pH 8.8	pH 9.0	pH 9.2

Bicarbonate indicator is red in equilibrium with atmospheric air. It changes from red to magenta to deep purple as carbon dioxide concentration decreases.
It changes from red to orange to yellow as carbon dioxide concentration increases

You are provided with the following equipment which you may use or not in your plan, as you wish.

- a supply of different day old mealworms
- syringes
- delivery tubes
- rubber tubing
- forceps
- bicarbonate indicator solution
- potassium hydroxide (to absorb atmospheric carbon dioxide)
- Vaseline petroleum jelly
- stopwatch
- colorimeter and cuvettes
- any normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.

Your plan should have a clear and helpful structure to include

- an explanation of the theory to support your practical procedure
- a description of the method used, including the scientific reasoning behind the method, a control experiment and any recommended safety measures
- an explanation of the dependent and independent variables involved
- relevant, clearly labelled diagram/s
- how you will record your results and ensure that they are as accurate and reliable as possible
- proposed layout of results tables and graphs with clear headings and labels
- the correct use of technical and scientific terms

[Total: 12]

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 **(a)** Compare replication in cells and the polymerase chain reaction (PCR). [8]

(b) Outline the basis and process of gel electrophoresis. [7]

(c) Describe the features of blood stem cells and explain their normal functions. [5]

[Total: 20]